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The efficacy of exergames-incorporated physical education lessons in influencing drivers of physical activity: A comparison of children and pre-adolescents

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ABSTRACT

Objective: This study examines the efficacy of incorporating exergaming into physical education (PE) lessons among children and pre-adolescents in influencing the social cognitive factors and behaviors of physical activity (PA), based on the theory of planned behavior (TPB).
Design: We conducted a six-week study centered on a 2 (exergaming: PE lesson with Wii vs. PE lesson without Wii) × 2 (age group: 10 years-old children vs. 12 years-old pre-adolescents) between-subject factorial design.
Method: 1112 participants were randomly assigned to the study conditions. At the end of the six-week program, participants completed a survey consisting of measures from the TPB variables.
Results: Exergaming significantly influenced PA attitude, subjective norm, intention, and strenuous exercise behavior, with participants in the Wii-incorporated PE lesson more likely to emerge with more positive beliefs and behaviors. Age significantly influenced outcome variables, with the effect of exergaming more pronounced among children than pre-adolescents in attitude, moderate and mild exercise behaviors.
Conclusions: Incorporating exergaming into PE lessons can be more effective than regular PE in enhancing PA beliefs and behaviors, particularly among younger children.
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Introduction

Overweight and obesity has been listed as the world's top fifth risk for mortality (World Health Organization, 2009). The world's prevalence of obesity has steadily risen, including that of childhood obesity. It is estimated that in 2010, 43 million children under five were overweight or obese (World Health Organization, 2009). In Asia, the prevalence of obesity among children under five has increased from 3.2% in 1990 to 4.9% in 2010 ("Child obesity growing in developing world: WHO," March 17, 2011). Similarly, the obesity rate among the Singapore population has grown from 6.9% to 10.8% in 2010 in just over a six-year period (Health Promotion Board, November 7, 2010). The obesity rate for school-going children in Singapore is comparable to that of the population at 10% in 2009 (Ministry of Education, November 24, 2010).

The rise in the obesity prevalence has been partly attributed to the declining amount of physical activity (Steinbeck, 2001), particularly as children got older (Nader, Bradley, Houts, McRitchie, & O'Brien, 2008). Recent attempts to encourage children to be more physically active have seen the inclusion of exergames, or active video games, into physical education (PE) lesson in schools (Davis, April 30, 2008; Henning, September 13, 2007; Lelchuk, February 20, 2007). Exergaming combines the features of video game play and physical activity on one platform (Sall & Grinter, 2007). Examples include the Wii and the Kinect systems. Players have to perform physical movements when engaging in exergames such as Dance Dance Revolution (DDR), Wii Sports, and Wii Fit. Previous research has found that a higher level of energy expenditure is generated when playing exergames than when undertaking sedentary activities (Graf, Pratt, Hester, & Short, 2009; Graves, Stratton, Ridgers, & Cable, 2008). Playing exergames has also resulted in increased motivation to play real sports (Klein & Simmers, 2009). Similarly, Brubaker (March 11, 2006) discovered that among overweight children, playing DDR at least five times per week resulted in them perceiving themselves to have better psychomotor coordination and stronger self-esteem. In turn, they were likely to start exercising more regularly. In addition, a significantly greater number of children found playing Wii Sports as more enjoyable than walking, watching television, and running (White, Kilding, & Schofield, 2009).

The premise of exergames in encouraging physical activity seems attractive. However, most of past exergaming studies have been conducted in Western countries with a relatively small sample (Peng, Lin, & Crouse, 2011). Hence, the present study aims to assess the efficacy of incorporating exergaming into PE classes in influencing the social cognitive factors of physical activity among

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more than 1000 children and pre-adolescents in Singapore. We utilize the theory of planned behavior (TPB; Ajzen, 1991) to guide the structure of our outcome variables and performance objectives. The TPB maintains that one's behavior is determined by the person's behavioral intention (plans or motivation to perform the behavior) and perceived behavioral control (perceived level of control in performing the behavior). Subsequently, intention is determined by attitude (positive or negative evaluation of the behavior), subjective norm (perceived social pressure to perform or not to perform the behavior) and perceived behavioral control.

As guided by the TPB, this research seeks to examine the impact of exergaming during PE lessons on participants' attitude, subjective norm, perceived behavioral control (PBC), intention, and behavior toward physical activity (mild, moderate, and strenuous exercise). The potentially differential impact of exergaming between children and pre-adolescents will also be assessed. We applied the intervention mapping approach principles (e.g., Bartholomew, Parcel, Kok, Gottlieb, & Fernandez, 2011) to help guide the design of our study in achieving the performance objectives that we set based on the TPB. We expect that the gaming nature of exergaming as well as rotating the games during the six-week intervention period to include both sports and dance game will inject an element of fun to the PE lessons, and hence, exergaming will increase participants' attitudes toward physical activity. For each game, participants were asked to play in a two-player mode such that social support, at least among their classmates, can be mobilized, and hence, supporting the attainment of the subjective norm objective. In addition, as the physical motions required from exergaming simulate the movements demanded from performing real sports, we expect that exergaming will enhance participants' confidence and PBC in performing physical activity and in their motor skills. Each class was also provided with 3 or 4 Wii consoles so that each participant could play more than once during each of the PE lesson. More game playing times, therefore, may result in more successful game plays, and subsequently a higher level of PBC.

Method

Participants

We conducted a six-week intervention study among 506 grade five ($M_{\text{age}} = 10.2$; SD = .55; 58% males) and 606 grade seven ($M_{\text{age}} = 12.2$; SD = .54; 51% males) students from three elementary schools and four secondary schools in Singapore. The racial composition of the participants consists of 63% Chinese, 24% Malays, 7% Indians, and 5% of other races. The study design and survey questionnaire were approved by Singapore's Ministry of Education. We also obtained written parental consent prior to the six-week intervention program.

Procedure

We deployed two independent variables, i.e., exergaming and age group, with each of the independent variables consisting of two levels. For exergaming, participants were assigned to undertake either PE lessons incorporated with exergaming or PE lessons following the regular curriculum without any exergaming element at all. As for the age group, the two levels were grade five children and grade seven pre-adolescents.

The students in grade five and seven undertook either one of the two conditions for six consecutive weeks, which are playing Wii active video games during PE lessons or undergoing the regular PE lessons with no exposure to Wii active video games. Each of the experimental conditions was randomly assigned to two to three classrooms of each school, depending on the school's cohort size. Participants in the Wii-incorporated PE lesson condition played Wii once a week during one of their weekly PE lessons. Each PE lesson was around 45–60 min long. In each school, three Wii video games consoles were set up and two players took turn to play at one Wii game console at any point of time. The size of each class ranged from around 30 to 35 students, and hence, each student was able to play for at least eight to ten minutes in each PE lesson. Three Wii active video games, namely DDR, Wii tennis, and Wii boxing, were chosen to be played in weekly rotation, such that each game was played twice in the six-week intervention program. The selection of these games is in line with past research which found that playing the beginner level of DDR fulfilled the minimal level for cardio respiratory requirement (Tan, Aziz, Chua, & Teh, 2002) and that playing Wii tennis and Wii boxing could result in energy expenditure level equivalent to that generated by moderate physical activity (Miyachi, Yamamoto, Ohkawara, & Tanaka, 2010). Participants in the PE lesson with no Wii exposure undertook the regular PE activities in accordance to each school's curriculum. At the end of the six-week program, participants were asked to complete a paper-and-pen survey consisting of measures from the TPB variables.

Measures

Except for the exercise behavior, all other items were measured on a five-point scale, with higher scores indicating higher levels of attitude, subjective norm, PBC, and intention.

Attitude

Participants were asked to respond to seven items on a five-point semantic-differential scale from Lowe, Eves, & Carroll's (2002) affective and instrumental attitude scale, such as *harmful–beneficial* and *enjoyable–unenjoyable* (Cronbach's $\alpha = .86$). The items were preceded by a statement "For me, exercising in my leisure time over the next 6 months would be ...."

Subjective norm

Four injunctive norm items such as "My friends would approve if I exercised regularly in the next 6 months" and two descriptive norm items such as "Most of my family members exercise regularly" were adapted from Rhodes and Courneya (2003). Responses ranged from strongly disagree at "1" to strongly agree at "5" (Cronbach's $\alpha = .79$).

Perceived behavioral control

Respondents were asked to respond to one item adapted from Armitage and Conner (2001) "For me, exercising regularly in my leisure time over the next 6 months will be .." on a five-point Likert scale *(extremely difficulty – extremely easy)*.

Intention

Three items were adapted from Armitage and Conner (2001) and Lwin (2009), such as "I intend to exercise regularly in my leisure time in the next 6 months". Responses ranged from strongly disagree at "1" to strongly agree at "5" (Cronbach's $\alpha = .83$).

Exercise behavior

Exercise behavior was measured using Godin and Shephard's (1985) leisure time exercise questionnaire in which participants were asked to indicate how many times on the average they perform strenuous, moderate, and mild exercises for more than 15 min during their free time on a typical 7-day period (a week).

Results

In order to assess if incorporating exergaming into PE classes and grade levels (being children and pre-adolescents) have a significant influence on physical activity beliefs and behaviors,

Table 1
ANOVA results of the effect of incorporating exergaming into PE lessons on physical activity beliefs and behaviors.

	PE lessons incorporating Wii ($n = 557$)		Regular PE lessons ($n = 555$)		F	Partial η^2
	M	SD	M	SD		
Attitude	4.11	.71	4.00	.70	13.67***	.012
Subjective norm	3.42	.79	3.33	.78	6.35*	.006
Perceived behavioral control	3.54	.99	3.56	1.01	.39	.000
Intention	3.83	.88	3.77	.87	3.71^	.003
Strenuous exercise	2.68	2.28	2.55	1.82	4.38*	.004
Moderate exercise	2.51	1.99	2.46	2.17	1.20	.001
Mild exercise	2.87	2.84	2.69	2.81	1.53	.001

Note. ^$p < .06$, *$p < .05$, ***$p < .001$.

a two-way analysis of variance (ANOVA) was performed. The results revealed that the main effect of exergaming was significant for attitude ($F(1, 1106) = 13.67$, $p < .001$, $\eta_p^2 = .01$), subjective norm ($F(1, 1104) = 6.35$, $p < .05$, $\eta_p^2 = .01$), strenuous exercise behavior ($F(1, 1108) = 4.38$, $p < .05$, $\eta_p^2 = .004$), and marginally significant for intention ($F(1, 1108) = 3.71$, $p = .054$, $\eta_p^2 = .003$). As shown in Table 1, those in the Wii-incorporated PE lesson were more likely to have a more positive physical activity attitude, subjective norm and intention, and undertake a higher level of strenuous exercise than those in the regular PE lesson. Similarly, the main effect of age or grade level was significant for attitude ($F(1, 1106) = 36.03$, $p < .001$, $\eta_p^2 = .03$), subjective norm ($F(1, 1104) = 10.37$, $p < .01$, $\eta_p^2 = .01$), PBC ($F(1, 1108) = 30.91$, $p < .001$, $\eta_p^2 = .03$), intention ($F(1, 1108) = 33.07$, $p < .001$, $\eta_p^2 = .03$), strenuous ($F(1, 1108) = 48.20$, $p < .001$, $\eta_p^2 = .04$) and moderate exercise behavior ($F(1, 1107) = 7.29$, $p < .01$, $\eta_p^2 = .01$). The results in Table 2 indicate that grade five children had a more favorable beliefs and behavior toward physical activity than grade seven pre-adolescents in all of the outcome variables, except for mild physical activity behavior. The main effects of both exergaming and grade level on mild exercise were not significant.

The interaction effect of both exergaming and grade level was found significant for three outcome variables, namely attitude toward physical activity ($F(1, 1106) = 8.08$, $p < .01$, $\eta_p^2 = .01$), moderate exercise ($F(1, 1107) = 10.15$, $p < .01$, $\eta_p^2 = .01$), and mild exercise behavior ($F(1, 1108) = 12.30$, $p < .001$, $\eta_p^2 = .01$), as shown in Fig. 1. Subsequently, we performed independent t-tests for the three variables to examine if the impact of incorporating exergaming into PE lessons is equal among children and pre-adolescents. Among children, we found that those having PE lessons with exergaming were more likely to have a higher level of attitude ($M_{\text{with Wii}} = 4.34$, $M_{\text{without Wii}} = 4.06$, $t(503) = 4.60$, $p < .001$), moderate exercise ($M_{\text{with Wii}} = 2.97$, $M_{\text{without Wii}} = 2.43$, $t(504) = 2.81$, $p < .01$), and mild exercise ($M_{\text{with Wii}} = 3.13$, $M_{\text{without Wii}} = 2.32$, $t(439.9) = 3.30$, $p < .01$) than those undertaking regular PE lessons. Among pre-adolescents, however, there was no significant difference in the three variables between participants in the two experimental conditions.

Discussion

Our findings revealed that incorporating exergaming into PE lessons was effective in enhancing the children's and pre-adolescents' levels of attitude and subjective norm toward physical activity, as well as strenuous exercise and, to a lesser extent, intention to exercise. Exergaming has been shown to provide players with an enjoyable exercise experience (Limperos, 2011) as they tend to view exergaming more as entertainment rather than exercise (Klein & Simmers, 2009). It is possible that the entertainment element of exergaming may have played a role in influencing participants' attitudes toward physical activity. In addition, the fact that students belonging to each class in the exergaming condition were all asked to play Wii in a multiplayer mode may have enhanced the strength of their group identification as the group shared the same experience of exercising through gaming, and consequently, this could have resulted in increased subjective norm. In line with the tenets of the TPB, the improvement in participants' attitude and subjective norm may have led to higher levels of physical activity intention and, subsequently, behavior. Furthermore, Klein and Simmers (2009) have found that playing exergames successfully may encourage children to engage in real sports. This corresponds to our results that exergaming enhanced participants' strenuous exercise behavior. These findings suggest that one of the ways in which schools can intervene in enhancing physical activity beliefs and behaviors among the young is through exergaming. Therefore, schools should consider incorporating exergaming into their PE curriculum, especially as the cost of Wii consoles is becoming cheaper.

In addition, our study found that even though the age gap of the two groups is only two years, the levels of physical activity beliefs and behaviors were generally lower among pre-adolescents than children. This is in line with past research which showed that the amount of physical activity declined as children got older (Nader et al., 2008). Our findings imply that health authorities and schools should focus their efforts on instilling positive beliefs and behaviors toward physical activity as early as possible as past research has shown that when

Table 2
ANOVA results of the effect of age on physical activity beliefs and behaviors.

	Children ($n = 506$)		Pre-adolescents ($n = 606$)		F	Partial η^2
	M	SD	M	SD		
Attitude	4.18	.68	3.95	.71	36.03***	.032
Subjective norm	3.45	.83	3.31	.75	10.37**	.009
Perceived behavioral control	3.73	1.00	3.40	.98	30.91***	.027
Intention	3.96	.84	3.67	.88	33.07***	.029
Strenuous exercise	3.06	2.33	2.24	1.72	48.20***	.042
Moderate exercise	2.66	2.15	2.34	2.01	7.29**	.007
Mild exercise	2.66	2.73	2.87	2.90	1.03	.001

Note. **$p < .01$, ***$p < .001$.

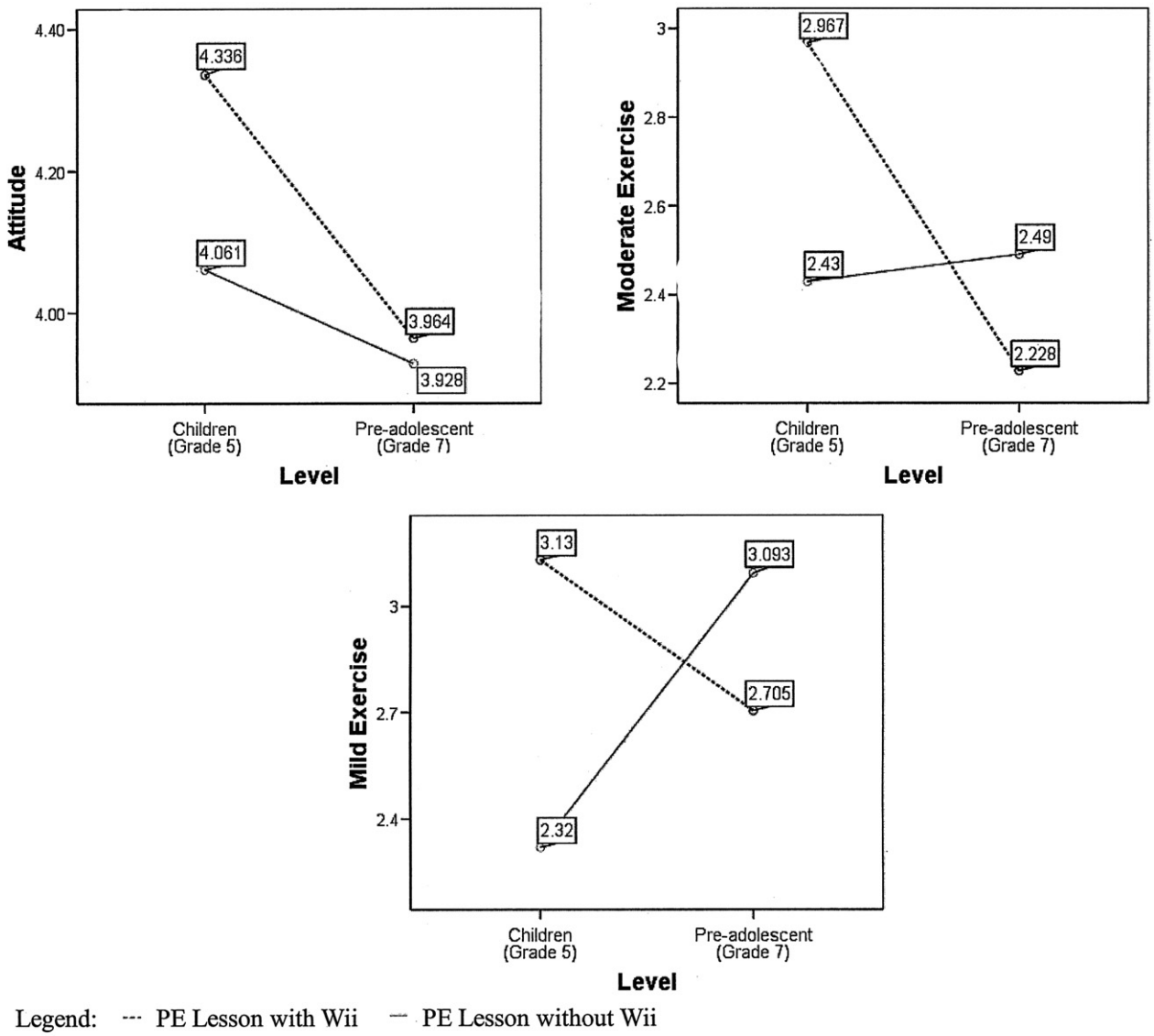

Fig. 1. The effect of exergaming and grade level on attitude toward physical activity, moderate exercise, and mild exercise.

children had developed positive social images of exercise earlier in their lives, they were more likely to have higher levels of athleticism and less likely to be obese (Hampson, Andrews, Peterson, & Duncan, 2007). Similarly, although the effect sizes were not large, our study suggests that it may be easier to effect change in the physical activity attitude and behavior among younger children as evidenced in our findings that the effect of exergaming is more pronounced in children than in pre-adolescents.

In summary, our findings show that incorporating exergaming into PE lessons can be an effective alternative to regular PE lessons in enhancing physical beliefs and behaviors, particularly among children. Future research can extend this study in several ways. First, a longitudinal study can be conducted to determine if the effect of incorporating exergaming in PE lessons on participants' physical activity beliefs and behaviors last up to six months or one year after the intervention. Second, future research can also assess the effectiveness of the different types of exergames, such as fitness-, adventure-, and dance-based games. Third, it has been found that successfully playing exergames could enhance the player's confidence (Klein & Simmers, 2009). Hence, subsequent study can investigate if a longer intervention period would result in higher levels of physical beliefs and physical activity. Finally, in comparing the physical activity beliefs and behavior across different age groups, future studies should attempt to compare groups with greater age difference to assess if a clearer pattern will emerge, such as between children aged 10 years-old and adolescents aged 15 or older.

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